

MA489 Interim Report

Machine Learning Models with Application of Radiation Therapy Quality Assurance

George Mandl

December 2023

1 Abstract

This research is motivated by a real world challenge – before radiation is conducted on a cancer patient, the radiation plan must pass a quality assurance (QA) testing process. It is assumed that the variability in a radiation plan has a large influence on the QA pass rates. There exists some metrics that measure the variation of a radiation plan. I have implemented variability metrics in Python and will pipe them into machine learning models like Neural Networks as predictor variables. I will attempt to predict QA pass rates using various machine learning models, with hopes of creating a tool that can be incorporated into the QA process of the research team at the Princess Margaret Hospital.

2 Background

When a patient is first diagnosed with cancer, it must be decided upon which therapy (or therapies) will be used to treat the cancer. The fundamental idea behind radiation therapy is that tumor cells are more sensitive to radiation than healthy cells. Ideally, the radiation is able to kill or significantly damage the tumors cells. If radiation therapy is pursued, then volumetric modulated arc therapy (VMAT) is a new and popular choice. The focus of my research in radiation therapy is limited to VMAT.

Before radiation can be applied to the patient there are several steps that need to be taken. First, imaging of the location of the tumor must be done and a plan of the dosage must be created. This plan needs to be tested in quality assurance (QA). It is only after QA, that the patient can receive the radiation therapy.

Radiation therapy is administered to the patient by a linear accelerator (linac). The linac rotates 360° around the treatment bench on which the patient lies. The gantry is the part of the linac which enables the rotation. The linac controls and shapes the radiation beam at various ‘control points’ throughout

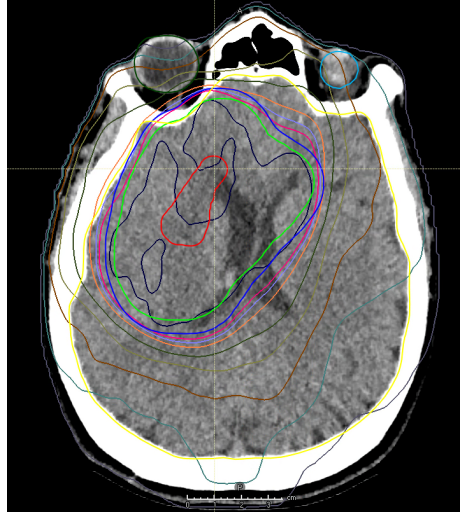


Figure 1: Treatment Planning Shapes.

the treatment. These control points occur at consistent angle intervals of the gantry's rotation.

The photon beam is shaped by the multi-leaf collimator (MLC). The leaves of a MLC are independently movable metal blades or leaves that can be positioned to create a variable opening for the photon beam to pass through, shaping the beam in by allowing or blocking specific areas to precisely target the treatment area while minimizing exposure to surrounding healthy tissues. The MLC has two leaf banks. In one of the leaf banks the leaves move from the left side into the isocentre of the photon beam and in the other leaf bank, the leaves move from the right. When the leaf in the left bank and the corresponding leaf in the right bank are at the same location there is no opening between them, meaning no radiation will pass through. Further, the distance between corresponding leaves corresponds to the amount of photon radiation being administered at a specific control point.

2.1 Treatment Planning

A treatment planner uses the imaging of the tumor to map the dose distribution of the VMAT. This is a large optimization problem in which the treatment planner looks to deliver a homogeneous dose to the tumor while minimizing the radiation that the surrounding organs receive. In Figure 1, there are various shapes that represent the tumor locations, the dose distributions and the surrounding organs. For example the right eye is mapped by a blue circle. Ideally the optimization would create a 3D continuous dose that is equally distributed at each angle but the delivery of this dosage is constrained by the linac. The motions of the leaves, the gantry angle motion, and the beam intensity deliv-

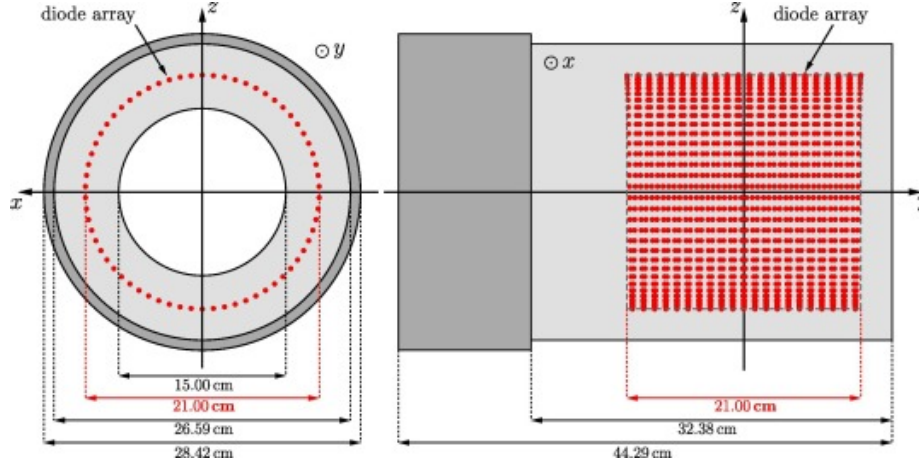


Figure 2: ARCCHECK Architecture. [Jiménez-Melguizo et al. 2020]

ered by the linac are all constraints that need to be considered in the planning phase. This is a highly complex process and it would be very irresponsible to immediately administer this dosage to the patient. Thus, the plan must undergo quality assurance to ensure that it is safe to be delivered to the patient.

2.2 Quality Assurance

In QA, the radiation plan is administered by the linac to a device called an ARCCHECK. ARCCHECK is a device that contains 1386 detectors which measure the radiation from the linac at various different points in space. Figure 2 shows an ARCCHECK devices and its detector structure. The success of QA is dependent on the QA pass rate. At each detector, a comparison is made between the measured dosage that the detector receives during QA and the dosage predicted by the treatment planner. If the difference between the two measurements is below a certain threshold then that detector passes the QA. The main pass rate is defined as the percentage of detectors that pass the dose comparison.

QA must be carried out for two reasons:

It is not possible to place detectors in or on the patient during the treatment so there is no way of determining the dosage the patient receives. The measured dose distribution from the QA is used as an approximation of the real dosage that will be received by the patient.

The synergy between the linac and the treatment plan must be evaluated. There are limitations to the motions of the linac's gantry angle and multi-leaf collimator. Testing must be done to determine if the linac can appropriately deliver the treatment plan devised by the planner.

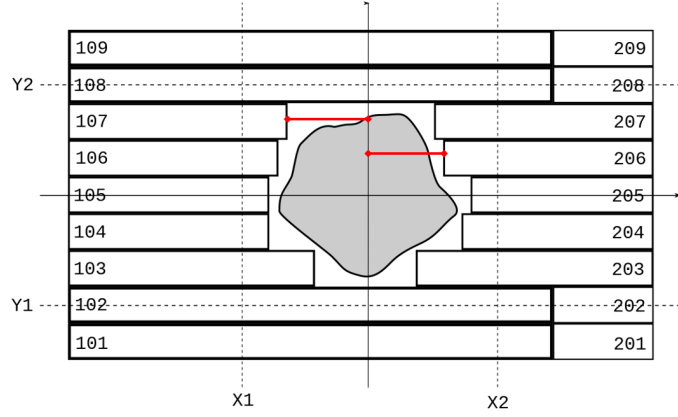


Figure 4: Depiction of Leaf Tip Positions
https://en.wikipedia.org/wiki/Multileaf_collimator

	1	3	5	7	9	11	13	15	17	19	...	341	343	345	347	349	351	353	355	357	359
0	-0.71	-0.56	-0.48	-0.48	-0.47	-0.44	-0.42	-0.67	-0.69	-0.46	...	0.29	-0.29	-0.41	-0.41	-0.42	-0.61	-0.61	-0.72	-0.75	-0.75
1	-0.71	-0.56	-0.48	-0.48	-0.47	-0.44	-0.42	-0.67	-0.69	-0.46	...	0.29	-0.29	-0.41	-0.41	-0.42	-0.61	-0.61	-0.72	-0.75	-0.75
2	-0.71	-0.56	-0.48	-0.48	-0.47	-0.44	-0.42	-0.67	-0.69	-0.46	...	0.29	-0.29	-0.41	-0.41	-0.42	-0.61	-0.61	-0.72	-0.75	-0.75
3	-0.71	-0.56	-0.48	-0.48	-0.47	-0.44	-0.42	-0.67	-0.69	-0.46	...	0.29	-0.29	-0.41	-0.41	-0.42	-0.61	-0.61	-0.72	-0.75	-0.75
4	-0.71	-0.56	-0.48	-0.48	-0.47	-0.44	-0.42	-0.67	-0.69	-0.46	...	0.29	-0.29	-0.41	-0.41	-0.42	-0.61	-0.61	-0.72	-0.75	-0.75
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
75	1.49	1.40	1.20	1.07	1.09	1.53	1.75	2.17	2.00	1.21	...	2.41	2.74	2.72	1.86	1.29	1.04	1.27	1.51	1.52	1.51
76	1.49	1.40	1.20	1.07	1.09	1.53	1.75	2.17	2.00	1.21	...	2.41	2.74	2.72	1.86	1.29	1.04	1.27	1.51	1.52	1.51
77	1.49	1.40	1.20	1.07	1.09	1.53	1.75	2.17	2.00	1.21	...	2.41	2.74	2.72	1.86	1.29	1.04	1.27	1.51	1.52	1.51
78	1.49	1.40	1.20	1.07	1.09	1.53	1.75	2.17	2.00	1.21	...	2.41	2.74	2.72	1.86	1.29	1.04	1.27	1.51	1.52	1.51
79	1.49	1.40	1.20	1.07	1.09	1.53	1.75	2.17	2.00	1.21	...	2.41	2.74	2.72	1.86	1.29	1.04	1.27	1.51	1.52	1.51

Figure 5: Matrix Representation of Leaf Tip Distances

80 leaves on each bank.

The nested structure of the JSON does not allow for very efficient and interpretable computations. Each computation requires several reference into various lists and dictionaries. The JSON files have been converted into matrices and vectors to reduce the complexity of computations and enhance the clarity of the code. An example of the matrices can be seen in Figure 5. In the matrices the columns represent the control points and the rows represent the leaf number. An entry in the matrix represents the specific leaf's tip positions at a specified control point of the treatment. The beam intensities can also be stored in a vector. Now, rather than representing the data which is a set of nested lists and dictionaries, the data is stored in a few matrices and vectors.

4 Objective

Ideally, it would be good to know before the QA process if a treatment plan will pass or fail. Essentially, I would like to leverage the geometric representation from the data to predict the QA pass rate. As expressed earlier, the pass rate is the proportion of detectors that pass the dosage comparison threshold. I will create a machine learning model for predicting the pass rates that, ideally, the hospital will be able to use between the treatment planning and QA processes. I also want to determine the factors that influence the pass rate. There exists various complexity metrics that are believed to have a negative relationship with the passing rate. I will introduce complexity metrics in Complexity Metrics. I would like to determine if there is a statistically significant relationship between complex metrics and the QA pass rate. If a relationship exists, measures can be taken to control for these metrics during the treatment planning or before the QA.

5 Complexity Metrics

Complexity metrics in VMAT (Volumetric Modulated Arc Therapy) refer to quantitative measures that assess the intricacy and sophistication of the treatment plan. These metrics evaluate various aspects of the treatment plan, providing insights into the plan’s efficiency and deliverability. It is believed that the more intricate and sophisticated a treatment is the more likely the plan will fail the QA. The following is an overview of the complexity metrics that I have implemented or am currently implementing.

5.1 Leaf Travel

Leaf travel is the distance the leaf travels over the entire rotation of the linac. It is computed by taking the sum of the differences between the leaf tip positions at one control point and the position at the previous control point, and can be written as the following equation:

$$LT = \sum_{i=1}^{n-1} |a_{leaf,i} - a_{leaf,i+1}| \quad (1)$$

where $a_{leaf,i}$ and $a_{leaf,i+1}$ are adjacent leaf tip positions and the leaf bank is irrelevant so long the leaves belong to the same one. Using the matrix representation, a vector of the leaf’s tip positions and a vector of its right neighbour’s tip positions are needed. These vectors are subtracted from each other and the absolute value is taken of the resulting vector (since distance is of interest, not displacement). Then, the sum of all the values in the resulting vector are taken and a final scalar is outputted, which is the leaf travel for one of the 160 leaves in the MLC. A higher leaf travel indicates more variability in the treatment plan. Figure 6 shows the leaf travel over the treatment plan. The blue line represents leaf bank A and the orange line represents leaf bank B.

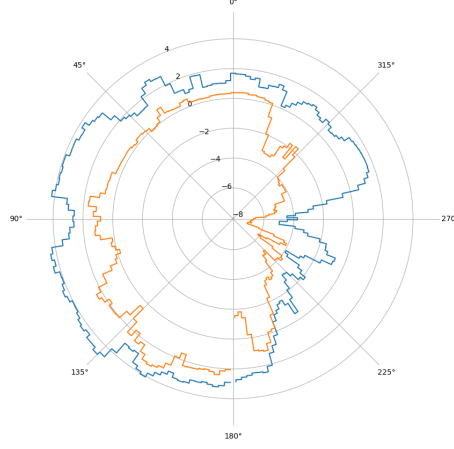


Figure 6: Leaf Travel Visualization. The blue line represents the leaf on the right bank and the orange line represents the leaf on the leaf bank.

5.2 Circumference over Area (CoA)

The area and circumference are separate computations that combine into the circumference over area complexity metric. In this application, area is the area of the opening in the MLC through which the photon beam will pass. The general area formula; length multiplied by width, is applied in this situation also. The width is the width of the leaf tip which is constant for all the leaves of the MLCs up until this point of the research. There are MLCs where the leaf tip width will vary and has been accounted for in the implementation. The length is represented by the distance between the leaf tips from the right bank and left bank. A visualization of segment area can be seen in Figure 7. The area calculation can be expressed by the following formula:

$$Area = \sum_{i=1}^n (a_{i,angle} - b_{i,angle}) * width \quad (2)$$

where $a_{i,angle}$ represent the value from the right leaf bank and $b_{i,angle}$ the value from left leaf bank of the same leaf number at a specific control point. Using matrix representation, the vectors of both leaf banks at a specified control point are needed. The left bank tip positions are subtracted from the right bank and these values are multiplied by the width of each leaf (which can be a scalar or a vector). Then the sum of all the values in the resulting vector are taken and a scalar is outputted. The scalar is the area for one of the 180 control points in the gantry angle.

Further, circumference is the circumference of the opening in the MLC through which the photon beam will pass. Circumference is calculated by adding all the leaf tip widths and the distances between neighbouring leaves. The visualization of segment circumference can be seen in Figure 8 and expressed as

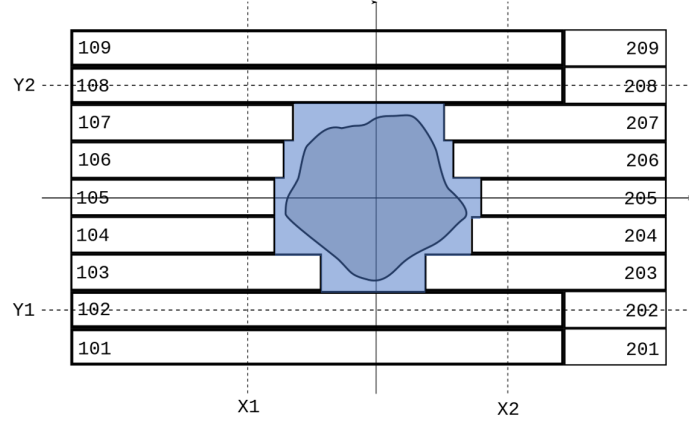


Figure 7: Segment Area Visualization
https://en.wikipedia.org/wiki/Multileaf_collimator

the following formula:

$$Circumference = \sum_{i=1}^{n-1} (|a_{i,angle} - a_{i+1,angle}| + |b_{i,angle} - b_{i+1,angle}| + 2 * width_i) \quad (3)$$

Where $a_{i,angle}$ and $a_{i+1,angle}$ are leaves in the same control point at adjacent leaf numbers and $a_{i,angle}$ and $b_{i,angle}$ are opposing leaves in the same control point and at the same leaf number. Using the matrix representation, the vectors of both leaf banks at a specified control point are needed. For both vectors, the absolute difference is taken between the leaf tip positions and the neighbouring leaf tip positions. The sum of these vectors and the widths of the leaves (which can be a scalar or a vector) is taken. Then the sum of all the values in the resulting vector is taken and outputs a scalar that represents the circumference for one of the 180 control points in the gantry angle.

The scalar representing the circumference is then divided by the scalar representing the area to determine the circumference over area (CoA). The idea behind CoA is that the higher the circumference is relative to the area, the higher the complexity of the shape through which the photon beam is passing through.

5.3 Radial Deviance

Radial deviance is a feature that is based on the ideal that a leaf that does not move (i.e. a constant leaf tip position) minimizes the complexity of the treatment field. The more the actual leaf tip position deviates from this ‘ideal’

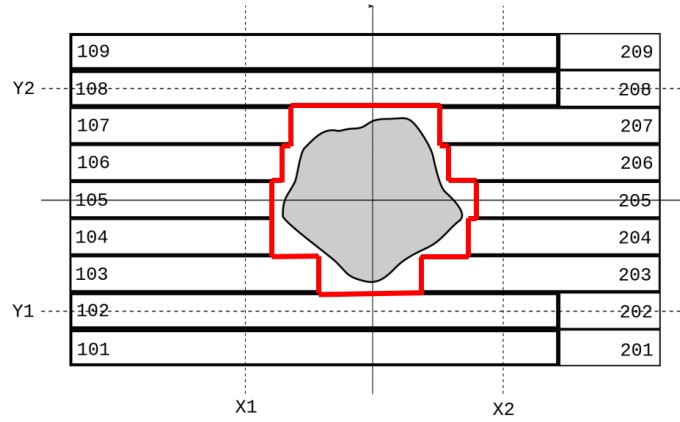


Figure 8: Segment Circumference Visualization,
https://en.wikipedia.org/wiki/Multileaf_collimator

constant position, the higher the complexity. Figure 10 shows an abstract visualization of this complexity metric. The implementation of radial deviance is still in process and a formula has yet to be developed.

5.4 Object-Oriented Programming

I would also like to highlight some computer science methodology that has been and will continue to be significant in the implementation of this research project in Python.

Object-oriented programming is a way of organizing and designing code by representing real-world entities as objects, which encapsulate data and the functions that operate on that data. The implementation of this research is rooted in a class of objects that which store data attributes and carry out functions on treatment fields. This ‘Field’ object reads the data from the JSON file, converts the data into matrices and vectors, and computes the complexity metrics. This object will be built upon with further complexity metrics and transformations to prepare for the machine learning models.

6 Future Scope

6.1 Predicting Pass Rates

Machine learning models will be implemented to predict the pass rates of the QA process. The input variables of the models will include the matrix and vector representation of the treatment plan, the complexity metrics calculated on the matrices and vectors of the treatment plan, and other information about

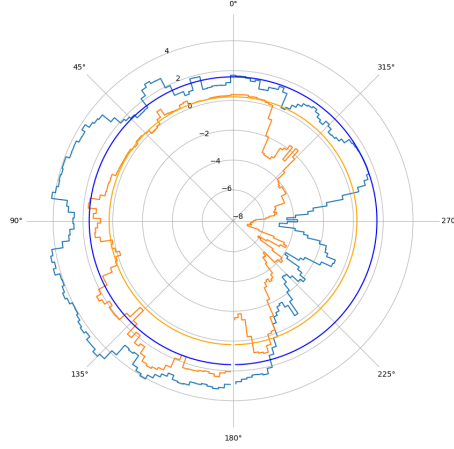


Figure 9: Radial Deviance Visualization. The blue line represents a leaf's tip positions from the right bank and the orange represents a leaf's tip positions from the left bank

the treatment such as which of the hospital's several linacs performed the treatment. Since the output variables (the pass rates) are factors constrained to the domain $[0,1]$, machine learning models that can map to probabilities need to be leveraged. These models include logistic regression, decision trees and their extensions, and neural networks. Recurrent neural networks (RNNs) are models that will definitely be investigated due to the correlation in the input data. Take, for example, the CoA of the control points in a treatment. The CoA at the first control point will likely be correlated to be CoA at the second control point because the leaves in MLC cannot move so quickly that the shape of the opening changes drastically. The feed-forward nature of RNNs will hopefully accommodate the correlation of the input.

6.2 Determining Statistically Significant Features

It would be beneficial to determine statistically significant features that can be engineered from the treatment plan and controlled for prior to QA. The complexity metrics introduced in the Complexity Metrics are potential candidates. In order to determine significance, the pass rates will be regressed on potential features using a multi-variable logistic regression. Using hypothesis testing, it can be determined which complexity metrics are statistically significant to passing rates and a framework can be built to control for these metrics between planning and QA.

7 Conclusion

In summary, this interim report provides a comprehensive overview of radiation therapy, particularly Volumetric Modulated Arc Therapy (VMAT). It emphasizes the critical steps in the treatment process, from imaging to radiation administration, using a linear accelerator. The report discusses the challenges in treatment planning and underscores the importance of Quality Assurance (QA) for safety. The research objective is well-defined: leveraging machine learning to predict QA pass rates and identify influencing factors. The methodology introduces complexity metrics and emphasizes Object-Oriented Programming for efficient code organization. The future scope outlines the plan to implement machine learning models and determine statistically significant features, showcasing a holistic approach that integrates medical insights with computational methodologies to enhance radiation therapy efficacy.

8 References

References

- [Elekta] Elekta (2011) *How a Linear Accelerator Works - HD*.
Retrieved from (<https://www.youtube.com/watch?v=jSgnWfbEx1A&t=376s>) on 11.12.23.
- [Jiménez-Melguizo et al. 2020] Miguel Jiménez-Melguizo, Miguel Espinosa, Joaquín Montes, Damián Guirado, Antonio M. Lallena (2020) *Response of the ArcCHECK device at 6MV and 15MV for VMAT and IMRT quality control Physica Medica* 80, 373 – 382.
(<https://www.sciencedirect.com/science/article/pii/S1120179720302945>)
- [Kamperis 2021] Stathis Kamperis (2021) *Complexity Analysis of Volumetric Modulated Arc Therapy Prostate Plans*.
Retrieved from (https://ekamperi.github.io/mrm_thesis/abstract.html) on 11.12.23.
- [Learn Oncology] Learn Oncology (2017) *How Radiotherapy Works!*
Retrieved from (<https://www.youtube.com/watch?v=ZgAi4wQLqZ0>) on 11.12.23.